

**DAY 3 – BIOS AND POST
TROUBLESHOOTING**

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Course: Diploma in Cyber Security

**Topic: BIOS and POST
Troubleshooting**

TASK 1 – COMMON BIOS / POST ERRORS

Error Message	What it Usually Means
CMOS Checksum Error	BIOS settings are reset or the CMOS battery may be weak.
Keyboard Not Found	The keyboard is not connected properly or there may be a connection problem.
No Boot Device Found	The computer cannot find the drive where the operating system is installed.
Memory Error	There may be a problem with the RAM or RAM connection.
CPU Fan Error	The CPU fan may not be connected properly or it may not be working.

TASK 2 – 3 BEEPS ON STARTUP

Problem:

When I start the PC, it gives 3 beeps again and again and there is no display on the screen.

Meaning:

For an AMI BIOS, 3 beeps can mean there is a problem with the RAM or memory.

What I would do:

1. Check the RAM

First, I would turn off the PC and remove the power. Then I would check if the RAM is properly fitted in the slot.

2. Check the RAM one by one

If the PC has two RAM sticks, I would test them one at a time. I would also remove and put the RAM back properly.

TASK 3 – BIOS SETUP SCREEN

What I have to do:

I need to open the **BIOS setup** on a PC or virtual machine and take a screenshot of the main BIOS screen.

Steps:

1. Restart the computer.
2. During startup, press the BIOS key such as **Delete, F2, F10 or Esc**.
3. Open the BIOS setup screen.
4. Find the **motherboard manufacturer**.
5. Find the **BIOS version**.
6. Take a screenshot.

Screenshot:

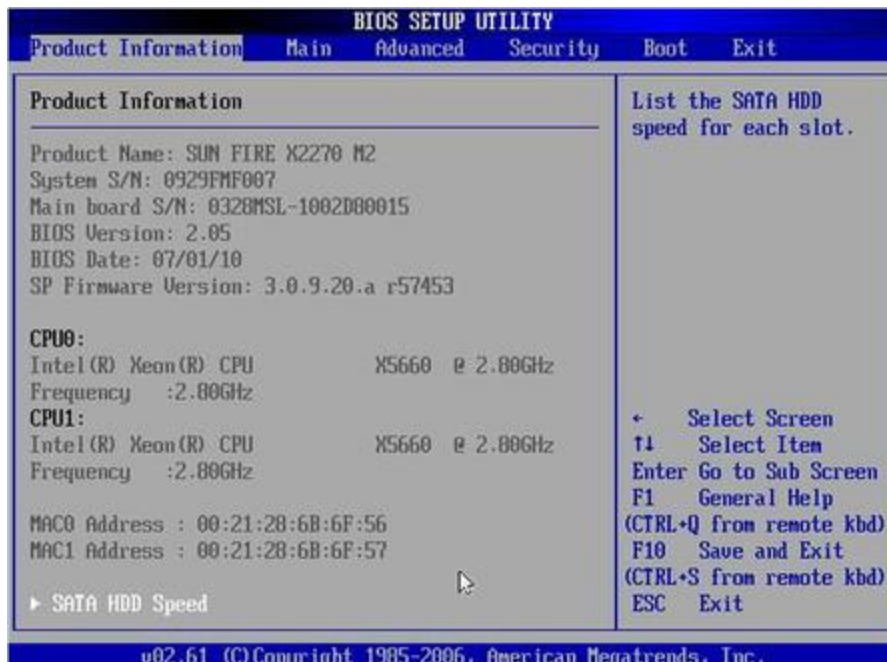


Figure 1: BIOS Setup Main Screen

Details:

Motherboard Manufacturer: Sun Microsystems / Oracle
(Product: SUN FIRE X2270 M2)

BIOS Version: 2.05 (Date: 07/01/10)

I will write the information that I can see on my BIOS screen.

TASK 4 – CMOS CHECKSUM ERROR

Error:

“CMOS checksum error – Defaults loaded.”

What does it mean?

This error means that the computer found a problem with the saved **BIOS settings**. Sometimes the BIOS settings can get reset.

One common reason can be a **weak or dead CMOS battery**.

How I would fix it:

1. Open BIOS

First, I would enter the BIOS setup and check the settings.

2. Check Date and Time

I would check if the date, time and other important settings are correct.

3. Check the CMOS Battery

If the same error comes again after restarting, I would check the CMOS battery.

4. Replace the Battery

If the battery is weak or dead, I would replace it with the correct compatible battery.

5. Save and Restart

After changing the settings or battery, I would save the BIOS settings and restart the PC to check if the error is fixed.

CONCLUSION

In this assignment, I learned about **BIOS and POST troubleshooting**.

I learned that BIOS and POST can help us understand some hardware problems when the computer starts. I also learned about common problems related to **RAM, keyboard, boot device, CPU fan and CMOS settings**.

I learned that **POST beep sounds can give us an idea about a hardware problem**. I also learned how to open the BIOS screen and find the **motherboard manufacturer and BIOS version**.

I also learned about the **CMOS checksum error** and how checking the BIOS settings and CMOS battery can help to fix this problem.

Overall, this assignment helped me understand the basic use of BIOS and POST and how they can help us find computer hardware problems.